Make Interactive Complex Heatmaps in R

Zuguang Gu[1,2]

[1]German Cancer Research Center, Computational Oncology, Molecular Diagnostics Program, National Center for Tumor Diseases (NCT), Im Neuenheimer Feld 280, 69120 Heidelberg, Germany
[2]DKFZ-HIPO (Heidelberg Center for Personalized Oncology), Im Neuenheimer Feld 280, 69120 Heidelberg, Germany

dkfz. GERMAN CANCER RESEARCH CENTER IN THE HELMHOLTZ ASSOCIATION

Research for a Life without Cancer

Introduction

Heatmap is a powerful visualization method on two-dimensional data to reveal patterns shared by subsets of rows and columns. I previously developed an R/Bioc package *ComplexHeatmap* that provides a general solution for making complex heatmaps. Here, I developed an R/Bioc package *InteractiveComplexHeatmap* that can easily export static complex heatmaps into interactive heatmaps as *Shiny* applications. It also provides flexible functionalities for integrating interactive heatmap widgets into other *Shiny* applications.

Features

1. It works for **all** kinds of heatmaps as long as they are generated by *ComplexHeatmap*.
2. It can also interactivate heatmaps from `heatmap()`, `heatmap.2()` and `pheatmap()`.
3. It allows integrating interactive heatmaps as parts of larger *Shiny* applications.
4. Response to mouse actions can be self-defined.

Usages

For the simplest usage, users only need to call one line of code after the heatmaps are generated.

```
htShiny() # yes, with no argument!
```

Examples

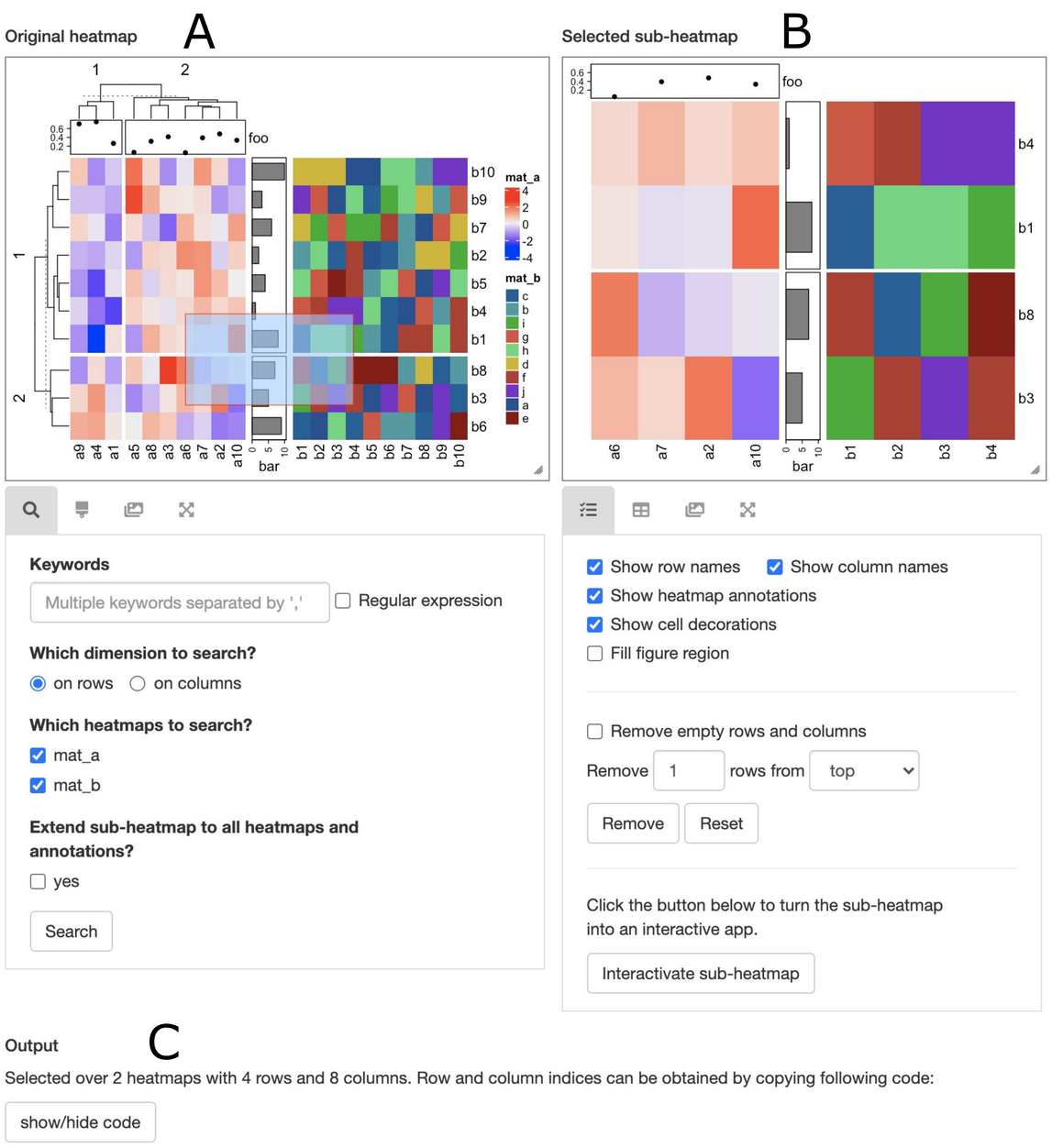

Figure 1. Default panels and tools in the interactive heatmap. A) The original heatmap. B) The selected sub-heatmap. C) The information of the heatmap cell that was clicked or the sub-heatmap that was selected.

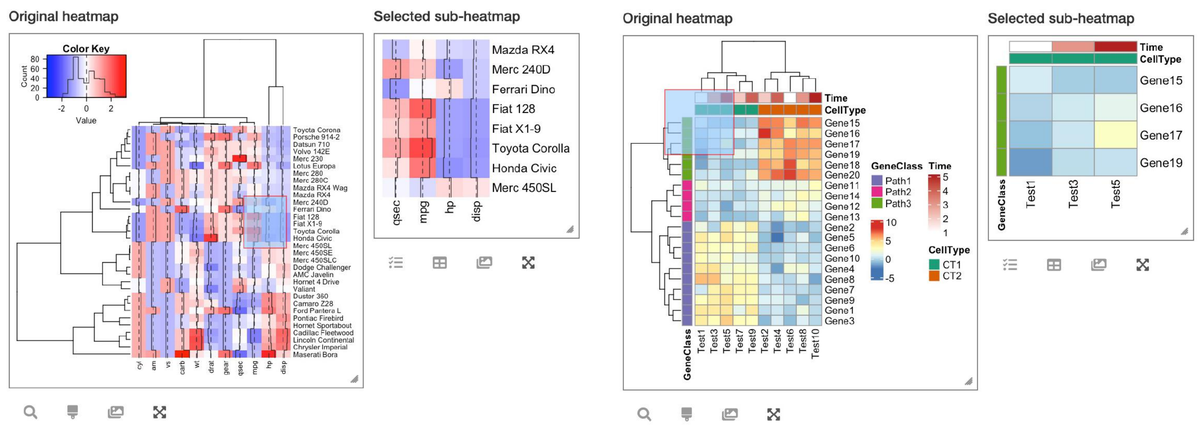

Figure 2. Interactive heatmaps from `heatmap.2()` and `pheatmap()`.

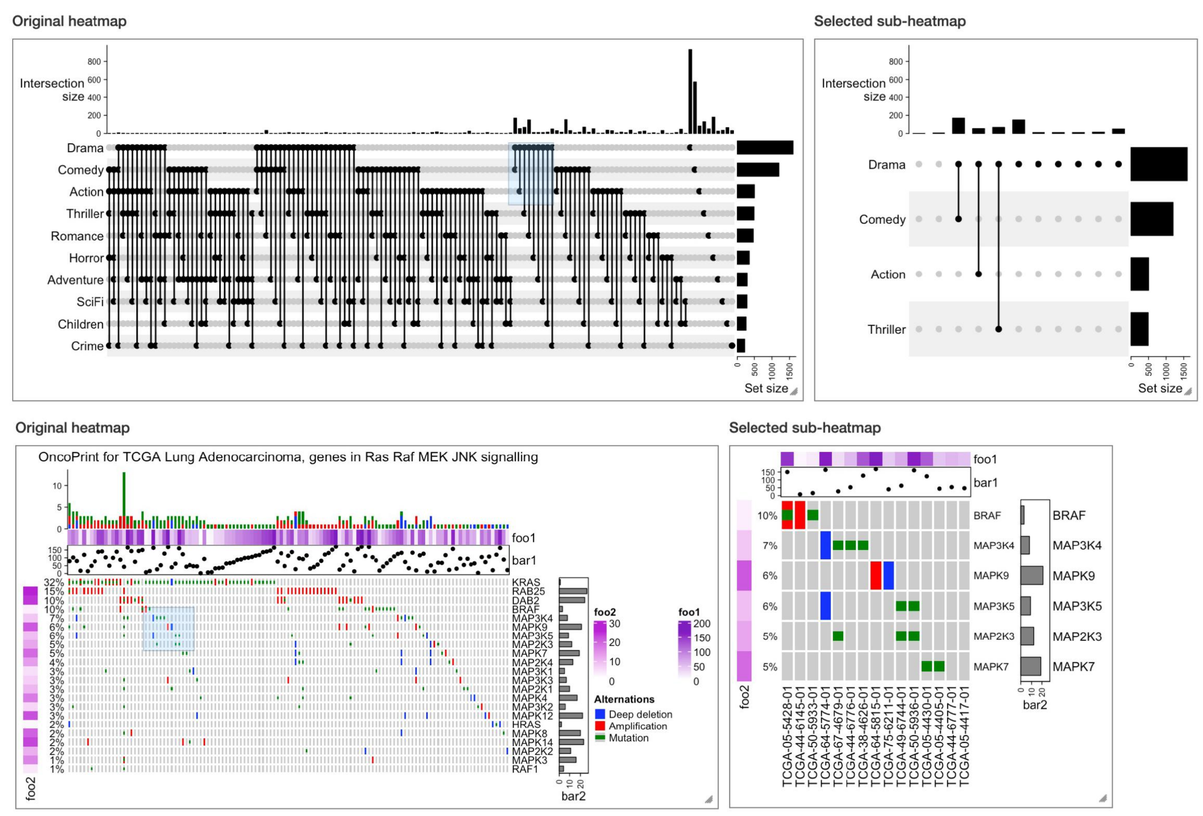

Figure 3. Interactive UpSet plot and oncoPrint.

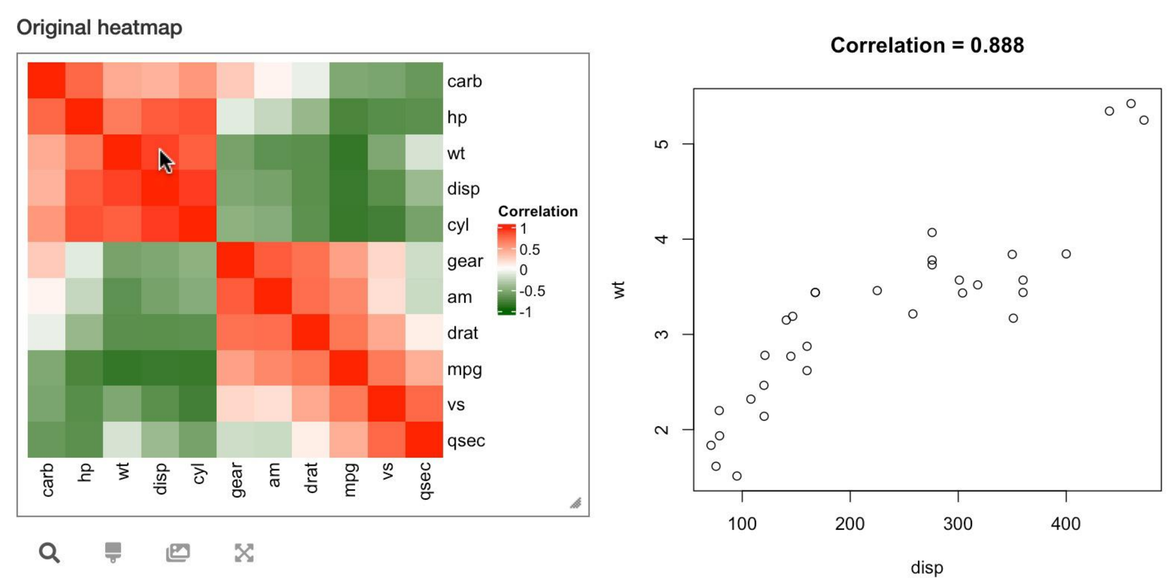

Figure 4. Self-define the response to the mouse click action.

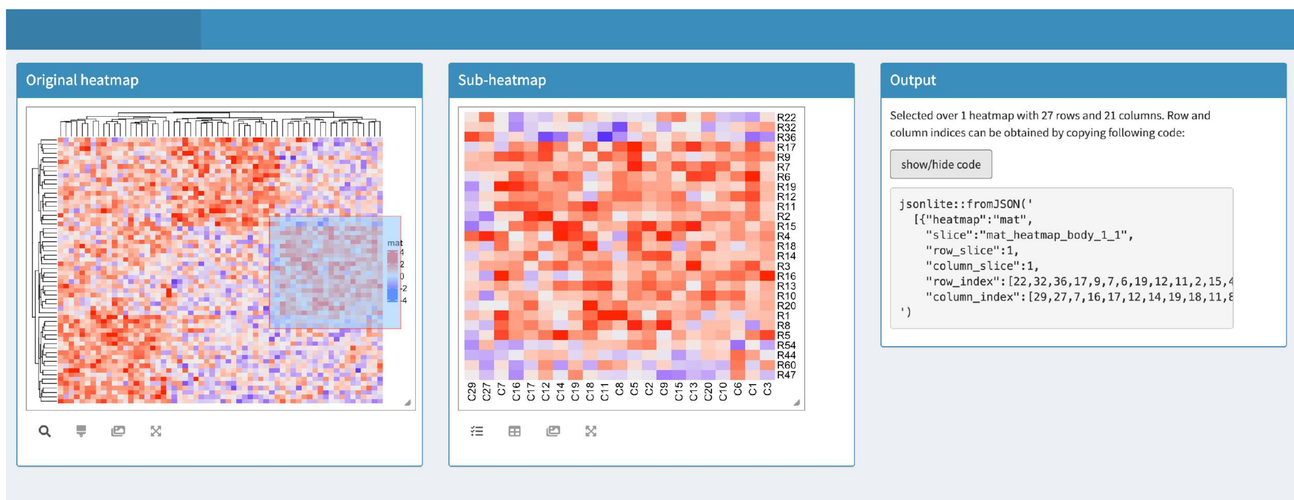

Figure 5. Integrate with *shinydashboard* pacakge.

GitHub: https://github.com/jokergoo, Twitter: @jokergoo_gu, bioRxiv: